Overall Progress :

The group worked together to design, develop, document, and refine the MacroTracker system by integrating all modules into a functional CLI application while applying OOP principles, performing debugging, and ensuring system usability and organization.

GROUP CONTRIBUTIONS:

Deoferio, Jan Zachary M.I.

1. A
2.

Mauro, Ryan Lewis B.

1. Final documentation organize
2. Code explanation, updates and error checking especially typo errors or missing code language.
3. Revised the 1.2 The project
4.

Mendoza, Nathaniel B.

1. Code sharing and updates in the group.
2. Lead developer , designed and implemented the overall system architecture for proper organization in order to integrate all modules into a functional python command line interface application.
3. Created the authentication system (Specifically the login/registration system with a proper approach to OOP to be specific inheritance so therefore it is a role based access control)
4. Approached OOP in a direct way meaning, properly addressed the 4 main parts of oop across the system.
5. Partial Flowchart for the Login system (Picture provided).
6. Partial Pseudocode for the system (Picture provided on the documentation below).
7. Created the food tracking module for macro tracking for tracking system for user input storing and computation of nutritional data.
8. Explored the applicable modules (built-in and external) for the command line interface project.
9. Organized the main folder and sub folder for the code and concluded separated .py files for easier programming.

10. Created dashboard, calendar, task management system for user interaction which these are used for displaying streaks, motivational messages, tracking of user activity and login, and also for deadline with completion of status.

11. Created menus for both user and admin.

12. Partial error handling, looking for possible errors and error catching.

13. The designer of the CLI, used an external module specifically “rich” and “pyfiglet” for a bigger font and colored display.

14. Database manager (Applied polymorphism) as recently discussed using a JSON-based database system.

15. Integrated / Connected all made modules into the main.py for usage.

Pilapil, Loreen Jae D.

1. Created the Readme file
2. Created the activity tracker part of the system
3. Revised the “Project” and “Specific Objective” part of the final documentation
4. Tweaked some parts of the final code as polishing
5. Explored the system thoroughly, acting as a user and also an administrator

Silao, Johrel Louie T.

1. I started editing the documentation file of our final project.
2. I inputted all the information into the documentation file that my groupmates and I have gathered so far.
3. Inputted the necessary information in the documentation file such as the General Problem and the Specific Problem.
4. Edited the documentation file in order for it to be clean and organized.
5. Revised the General Problem and Specific Problem.
6. Inputted information at the Solution section.
7. Inputted data at the General Objectives.
8. Inputted data at the introduction of the Flowchart.
9. Organized the documentation paper.

Overall Documentation:

MacroTracker: A Macronutrient Intake Tracker that Provides a Guideline Based on their Body Mass Index

Deoferio, Jan Zachary M.I.
Mauro, Ryan Lewis M.I.
Mendoza, Nathaniel B.
Pilapil, Loreen Jae D.
Silao, Johrel Louie T.

Technological Institute of the Philippines
Quezon City

April 2026

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PROBLEM AND ITS BACKGROUND

1.1 Introduction

The struggles of individuals maintaining their healthy food intake and monitoring their body mass index in the modern world.There is a specific requirement for energy intake that an individual's body needs in order for it to function properly. Another is to ensure their physical growth and prevent any diseases that they may encounter. However, many people struggle to monitor their own macronutrients - carbohydrates, proteins, and lipids, which are essential for primary energy production and bodily functions.

Manual tracking difficulties are one of the problems that we researchers observed. With this, individuals lack the ability to record their meals and log their daily physical activities. Maintaining a good health and wellness, the demand for high-end technologies that supports and monitors the macronutrient intake that an individual needs has spikes, as well as promoting healthy lifestyle activities or routines. Another limitation is that there is a need for personalized health journeys that calculate the person's BMI (Body Mass Index) in order to provide a suggestion/s on what meals and physical activities that they should perform daily.

To address these problems, our project aims to create a code-based (CLI) macronutrient intake monitor that shows a guideline on what kind of physical activities they should do and what dietary foods they should take that is based on their BMI. The program will be designed to record, track and provide automated exercise and diet suggestions efficiently. Using the VScode Python IDE, an individual's health journey can be tracked and monitored digitally which is efficient for users.

1.2 The Project

The MacroTracker is a command-line health monitoring system that tracks an individual's daily macronutrient intake. The system allows users to log meals, record physical activities, and monitor their nutrient intake accurately. The system also computes the user's Body-Mass Index (BMI) to help generate a personalized diet and exercise recommendations. The system has user authentication features, including sign up and log in to ensure security. An admin log in is also available to keep sensitive system operations. Overall, MacroTracker serves as a tool that supports the users to make informed decisions about their health and nutrition.

1.3 Objectives

<General Objectives>

The project aims to aid or support individuals who are struggling monitoring and tracking their macronutrient intake and provide a good healthy journey. The program allows the users to sign up for new users by providing a username and password. Once the users have signed up, they can now view their own tracker and they would be able to protect themselves from any sensitive actions. With this, the users can now monitor and track their macronutrient intake and the system would provide a suggestion on what the user should exercise and have a proper diet meal that is based on their BMI.

<Specific Objectives>

- 1. Develop a CLI system using Python
- 2. Create a sign up option for new users
- 3. Implement a user log in to access personalized data
- 4. Implement admin log in with password to protect sensitive data
- 5. Compute Body Mass Index (BMI)
- 6. Track users macronutrient intake and physical activities
- 7. Automate diet and exercise recommendations based on computed BMI

1.4 Flowchart of the System

This flowchart demonstrates the logic and operational structure of the user interface and its authentication as well as the registration system. This serves as a visual representation for the program's decision making path/s.

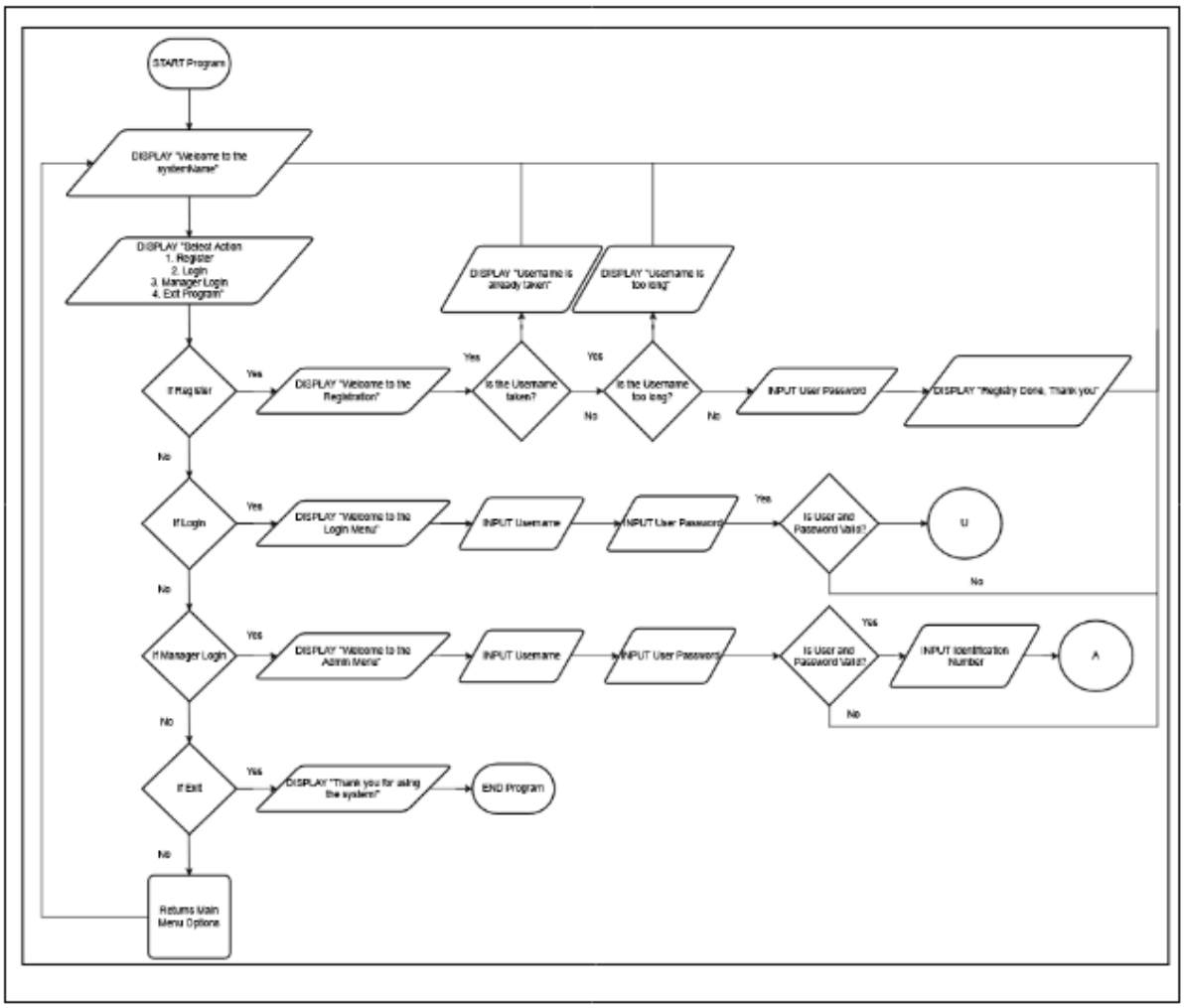


Figure 1: CRUD Application Flow Chart

Flow Chart Description

Figure 1 Illustrates the application of Flow of how the system will work using Create, Read, Update and Delete.

Legend:

A - Admin

U- User

IMPLEMENTATION OF OOP

2.1 Object Oriented Programming Implementation

<Subsection introduction>

2.1.1 Encapsulation

2.1.2 Inheritance

2.1.3 Abstraction

2.1.4 Polymorphism

<Implementation of OOP (Encapsulation,Inheritance, Abstraction, Polymorphism>

Figure 3. <name of figure>

Code discussion

2.2 Python Elements (Python Modules, PIP)

<Consider having a table here>

Module	Uses and Purpose
os	Clears terminal screen every menu for cleanliness.
time	Handle timestamps for logs
random	Random number generation used for admin id and colors for the display text macrotracker
json	Polymorphism (used to store and retrieve user data in JSON format)
abc	To use abstraction
rich	Colored Text Output
pyfiglet	To generate ASCII art for macro tracker title display
questionary	Interactive CLI menus for user input

RESULTS AND DISCUSSION

3.1 Output

<Subsection introduction >

<Output + discussion of the Figure>

GUI or CLI

Figure 4. <name of figure>

3.2 Conclusion

<conclude results and discussions, add recommendations>

REFERENCES

Dubey, Viksar, and Jérémie Lumbroso. "Analyzing the Effects of Caloric and Macronutrient Consumption on Productivity: A Quantified Self Approach to Assess Work Patterns and Habits." Journal of Student Research, vol. 12, no. 4, 30 Nov. 2023, pdfs.semanticscholar.org/5737/a578ec0b3d790e525aa198d9a34c52e1a644.pdf, https://doi.org/10.47611/jsrhs.v12i4.5729. Accessed 21 Mar. 2025.

Savarino, Giovanni, et al. "Macronutrient Balance and Micronutrient Amounts through Growth and Development." *Italian Journal of Pediatrics*, vol. 47, no. 1, 8 May 2021, www.ncbi.nlm.nih.gov/pmc/articles/PMC8106138/.

Selvaraju, T., Dakshinamurthi, V., Badurudeen, G., Prabhakaran, N., Gopi, P., & Hussain, M. A. N. (2025). Food detection and estimation of calories and other macro-nutrients and features to support healthy lifestyle. AIP Conference Proceedings, 3279, 020129. https://doi.org/10.1063/5.0263063

APPENDICES

Appendix A: Github Repo

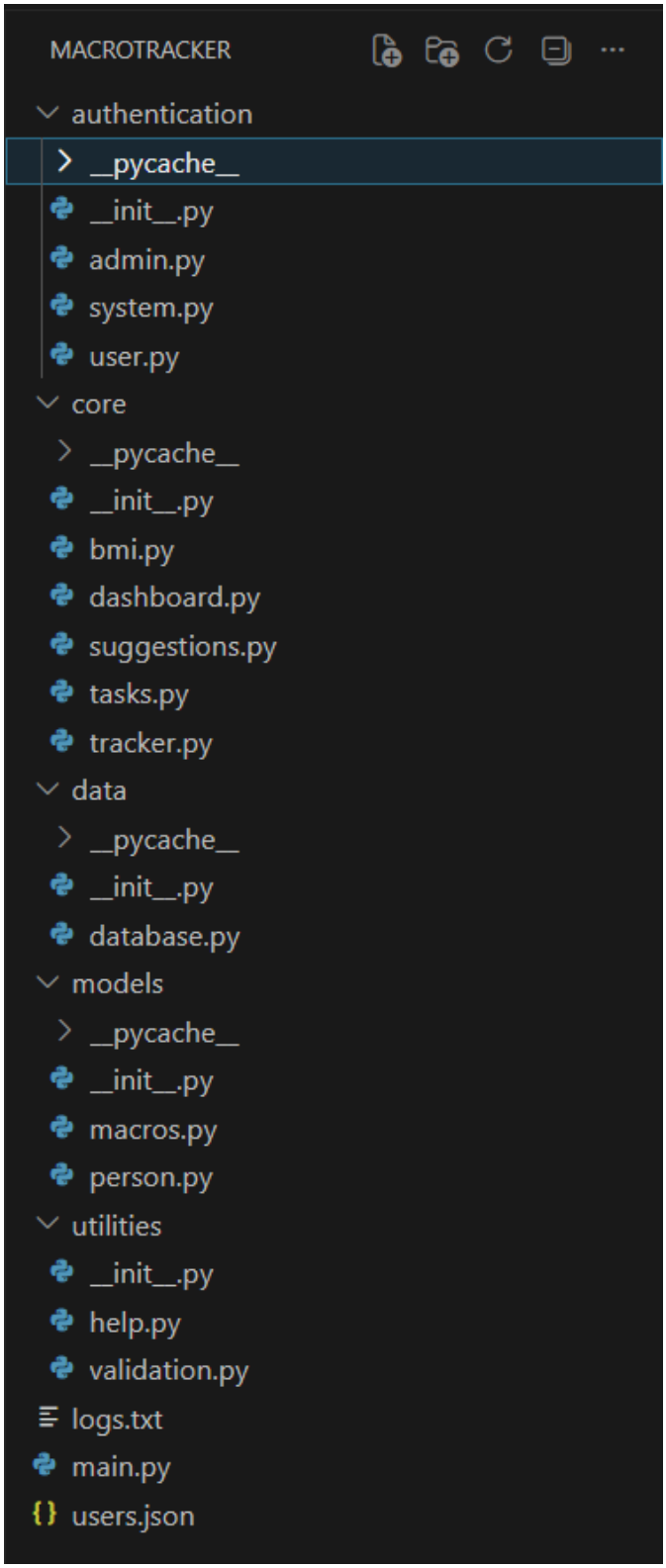
https://github.com/NatMendoza/CPE009A-FINAL-PROJECT-OOP1/tree/main

Appendix B: Youtube Link/Video Demo

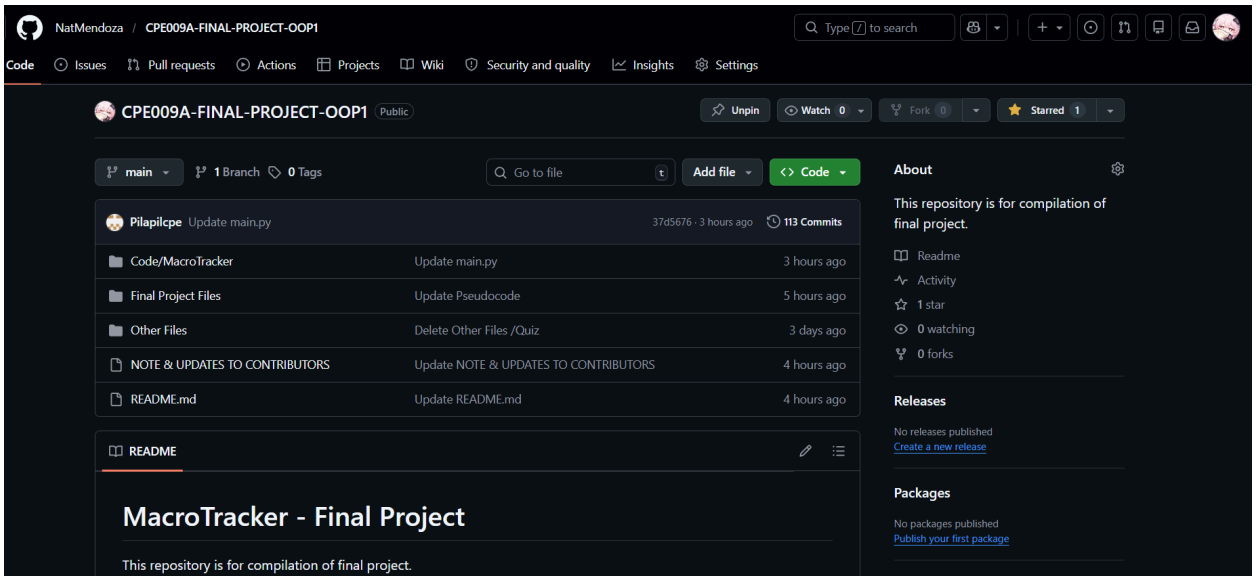
GITHUB REPOSITORY:

https://github.com/NatMendoza/CPE009A-FINAL-PROJECT-OOP1/tree/main

CODE STRUCTURE:

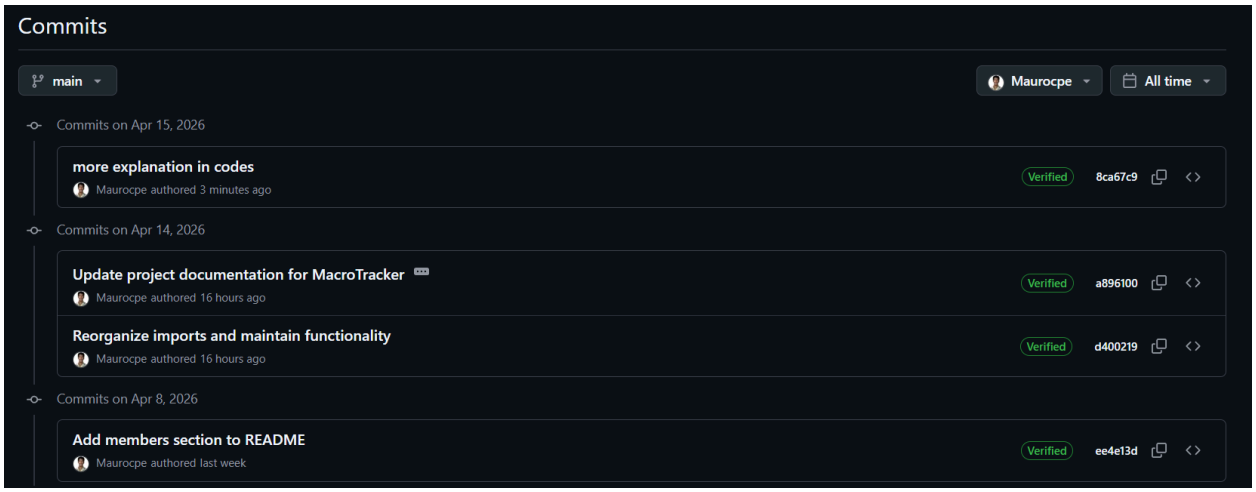


GITHUB CONTRIBUTIONS PROOF: (110+ Github Commits so far).

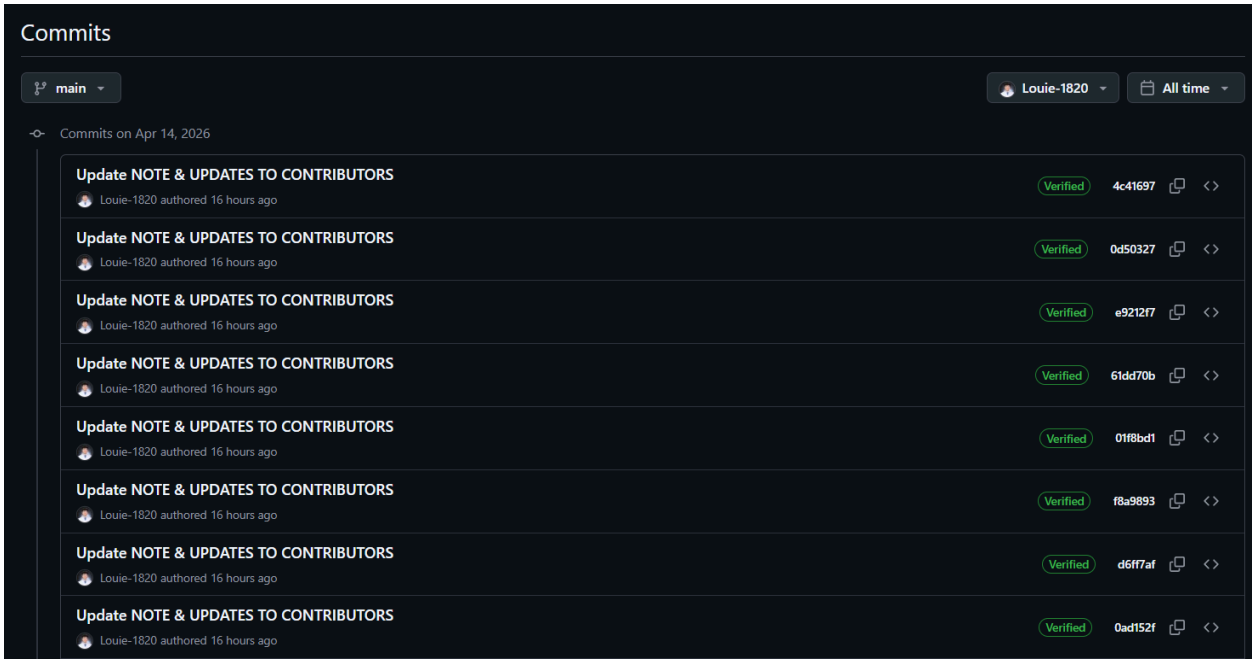


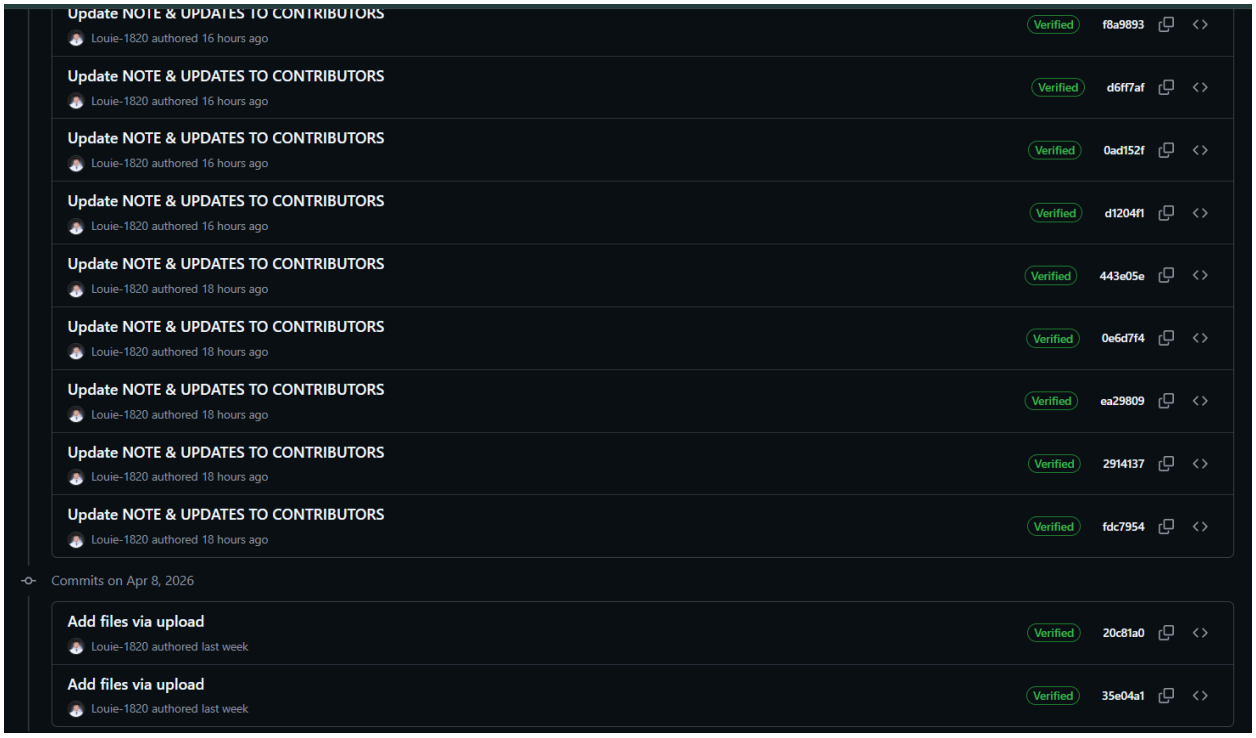
LIST OF GUTHUB COMMITS (PROOF):

Member #1



Member #2





Member #3

Commits

main | Pilapilcpe | All time

Commits on Apr 15, 2026

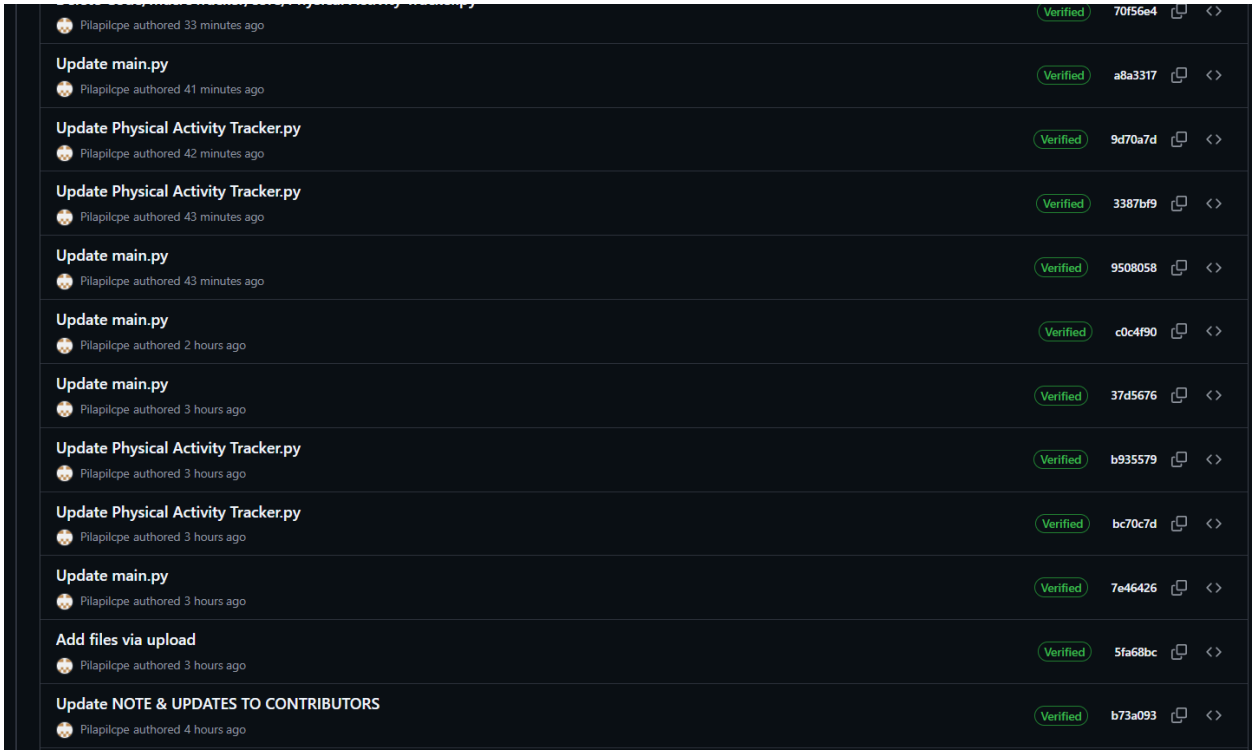
Update README.md Pilapilcpe authored now	Verified	f8685f7
Update README.md Pilapilcpe authored now	Verified	9cae644
Update main.py Pilapilcpe authored 28 minutes ago	Verified	6ee0229
Add files via upload Pilapilcpe authored 30 minutes ago	Verified	1d9f324
Delete Code/MacroTracker/core/Physical Activity Tracker.py Pilapilcpe authored 33 minutes ago	Verified	70f56e4
Update main.py Pilapilcpe authored 40 minutes ago	Verified	a8a3317
Update Physical Activity Tracker.py Pilapilcpe authored 41 minutes ago	Verified	9d70a7d
Update Physical Activity Tracker.py Pilapilcpe authored 3 hours ago	Verified	3387bf9
Update NOTE & UPDATES TO CONTRIBUTORS Pilapilcpe authored 4 hours ago	Verified	b73a093
Update main.py Pilapilcpe authored 4 hours ago	Verified	3a17746
Update README.md Pilapilcpe authored 4 hours ago	Verified	f016311
Update README.md Pilapilcpe authored 4 hours ago	Verified	9091d0d
Update README.md Pilapilcpe authored 4 hours ago	Verified	512138b
Update README.md Pilapilcpe authored 4 hours ago	Verified	def162c
Update NOTE & UPDATES TO CONTRIBUTORS Pilapilcpe authored 6 hours ago	Verified	e819a27

Commits on Apr 14, 2026

Update NOTE & UPDATES TO CONTRIBUTORS Pilapilcpe authored 17 hours ago	Verified	0c49ea5

Commits on Apr 12, 2026

Update CPE009A Final Documentation Pilapilcpe authored 3 days ago	Verified	fde3e1d


Pilapilcpe authored 33 minutes ago
Verified
70f56e4
Update main.py
Pilapilcpe authored 41 minutes ago
Verified
a8a3317
Update Physical Activity Tracker.py
Pilapilcpe authored 42 minutes ago
Verified
9d70a7d
Update Physical Activity Tracker.py
Pilapilcpe authored 43 minutes ago
Verified
3387bf9
Update main.py
Pilapilcpe authored 43 minutes ago
Verified
9508058
Update main.py
Pilapilcpe authored 2 hours ago
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c0c4f90
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Update Physical Activity Tracker.py
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Verified
bc70c7d
Update main.py
Pilapilcpe authored 3 hours ago
Verified
7e46426
Add files via upload
Pilapilcpe authored 3 hours ago
Verified
5fa68bc
Update NOTE & UPDATES TO CONTRIBUTORS
Pilapilcpe authored 4 hours ago
Verified
b73a093

Member #4

Commits

main | NatMendoza | All time

Commits on Apr 15, 2026

Update Pseudocode NatMendoza authored 5 hours ago	Verified	36c8998
Update Other Pseudo NatMendoza authored 5 hours ago	Verified	7e8d771
Update dashboard.py NatMendoza authored 5 hours ago	Verified	4d9294a
Created tasks.py NatMendoza authored 5 hours ago	Verified	846abf7
Create dashboard.py NatMendoza authored 5 hours ago	Verified	4a21e7b
Big update to the database.py NatMendoza authored 5 hours ago	Verified	17c417c
Slight changes to tracker.py NatMendoza authored 5 hours ago	Verified	d31f4e8

Commits on Apr 13, 2026

Update system.py NatMendoza authored 2 days ago	Verified	2011114
Update main.py NatMendoza authored 2 days ago	Verified	9bbda13
Update main.py NatMendoza authored 2 days ago	Verified	52946b6
Update NOTE & UPDATES TO CONTRIBUTORS NatMendoza authored 2 days ago	Verified	24329de
Update main.py NatMendoza authored 2 days ago	Verified	77cf8ea
Update main.py NatMendoza authored 2 days ago	Verified	c4259d2
Create __init__.py NatMendoza authored 2 days ago	Verified	3ebcb2a
Update person.py NatMendoza authored 2 days ago	Verified	1587222
Update macros.py NatMendoza authored 2 days ago	Verified	a05cb5c
Update database.py NatMendoza authored 2 days ago	Verified	39027c5
Update NOTE & UPDATES TO CONTRIBUTORS	Verified	45c3e41

NatMendoza authored 2 days ago

Update bmi.py NatMendoza authored 2 days ago	Verified	fefaa0e
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Update user.py NatMendoza authored 2 days ago	Verified	34d5f8f
Update system.py NatMendoza authored 2 days ago	Verified	613531e
Update admin.py NatMendoza authored 2 days ago	Verified	805e506

Commits on Apr 12, 2026

Create HOW SHOULD THE SYSTEM WILL WORK NatMendoza authored 3 days ago	Verified	18b653f
Rename ADMIN "Secret" Code to ADMIN "Secret" Code or Password NatMendoza authored 3 days ago	Verified	c7bd30d
Create ADMIN "Secret" Code NatMendoza authored 3 days ago	Verified	aa48c5a
Update person.py NatMendoza authored 3 days ago	Verified	f6eeddc

Commits on Apr 12, 2026

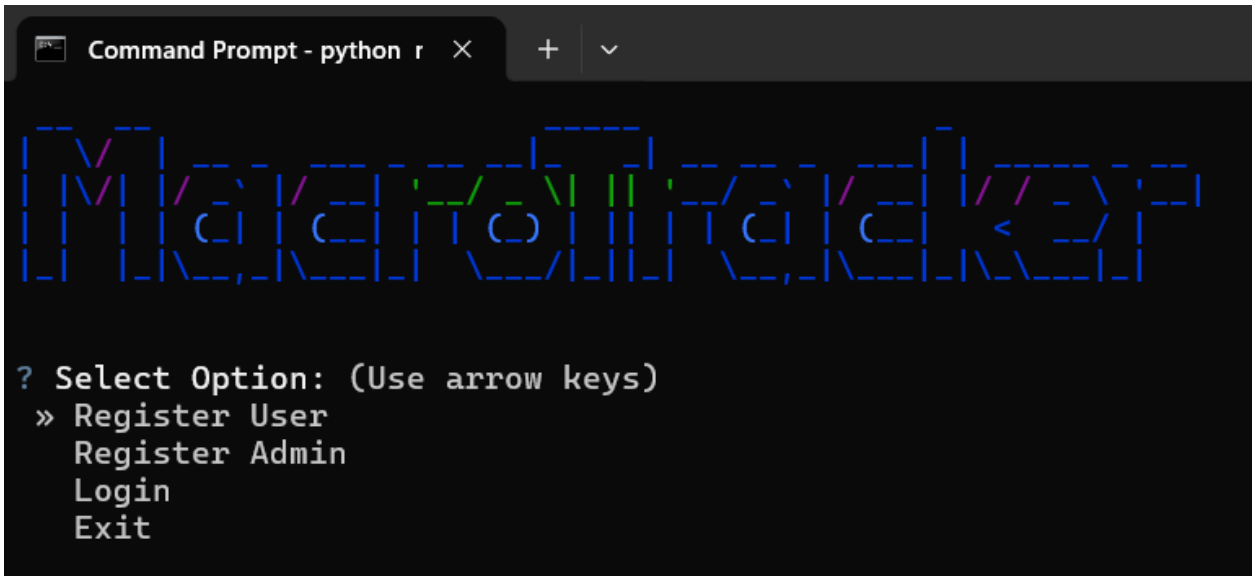
Update admin.py NatMendoza authored 3 days ago	Verified	5288027
Create Quiz NatMendoza authored 3 days ago	Verified	622b755
Delete Quiz Progression NatMendoza authored 3 days ago	Verified	5d90fb2
Create NatMendoza authored 3 days ago	Verified	ef2e61d
Delete Code directory NatMendoza authored 3 days ago	Verified	3cd18d9
Add files via upload NatMendoza authored 3 days ago	Verified	8a9f620
Create Overall NatMendoza authored 3 days ago	Verified	f4cc5a7
Delete Code Compilation NatMendoza authored 3 days ago	Verified	2217ebb
Create Code Compilation NatMendoza authored 3 days ago	Verified	a4fa326

Commits on Apr 8, 2026

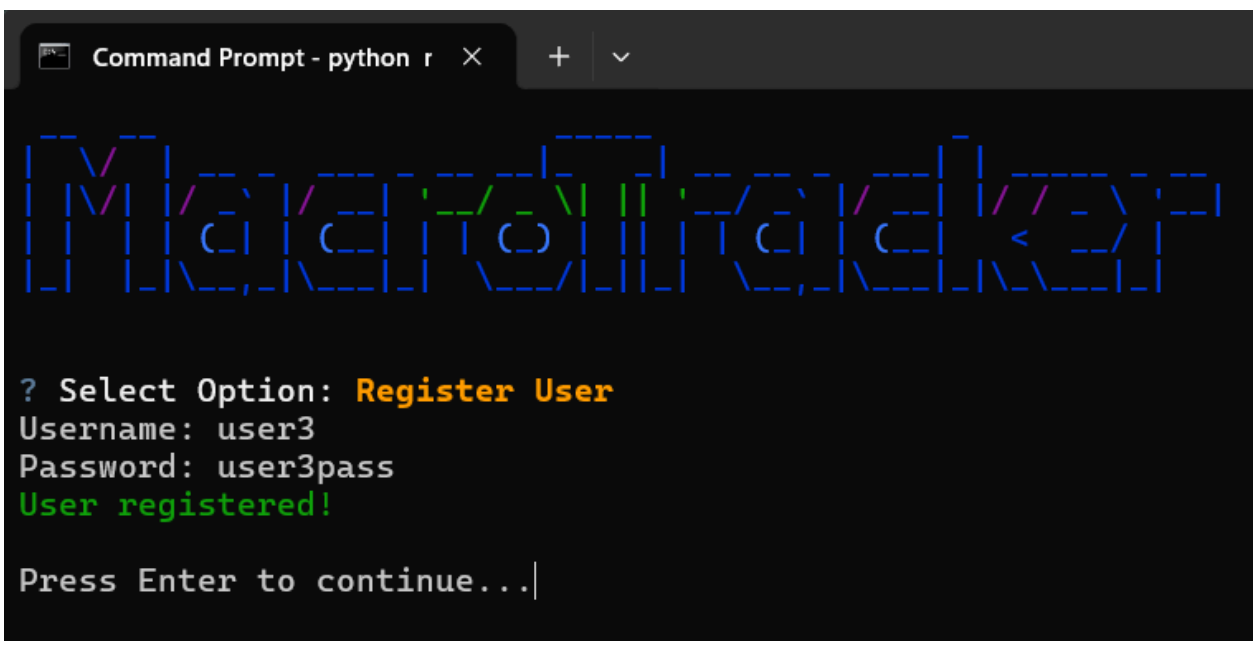
Initial commit NatMendoza authored last week	Verified	04965e3

PARTIAL PROJECT OUTPUT:

1. **First Look (Main Menu)**



2. **Registration User**



3. **Registration Admin**

```
 __  __                    _____                  _
|  \/  | __ _  ___ _ __ __|_   _| __ __ _  ___| | _____ _ __
| |\/| |/ _` |/ __| '__/ _ \| || '__/ _` |/ __| |/ / _ \ '__|
| |  | | (_| | (__| | | (_) | || | | (_| | (__|   <  __/ |
|_|  |_|\__,_|\___|_|  \___/|_||_|  \__,_|\___|_|\_\___|_|


? Select Option: Register Admin
Enter Admin Secret Code: MacroTrackerAdmin
Admin Username: NatMendoza
Password: NatMendozaPassword
Admin created! ID: 670933

Press Enter to continue...
```

4. **User Menu**

```
 __  __                    _____                  _
|  \/  | __ _  ___ _ __ __|_   _| __ __ _  ___| | _____ _ __
| |\/| |/ _` |/ __| '__/ _ \| || '__/ _` |/ __| |/ / _ \ '__|
| |  | | (_| | (__| | | (_) | || | | (_| | (__|   <  __/ |
|_|  |_|\__,_|\___|_|  \___/|_||_|  \__,_|\___|_|\_\___|_|


? Welcome user3 (Use arrow keys)
 » Dashboard
   Calendar
   Add Food
   View History
   Delete Food Entry
   Calculate Total
   BMI Check
   Add Task
   View Tasks
   Complete Task
   Delete Account
   Logout
```

5. **Admin Menu**

Note: *Didn't have time to send the inside of each menu.*